Two spins coupled to an XXZ bulk

Reduced dynamics and quantum-state texture

Spin transport project

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Abstract

We study the joint reduced dynamics of two spin-1/2 probes, each attached to one end of a finite XXZ bulk. The total Hamiltonian generates unitary evolution, and the initial probe–bulk state is a product. For Ising boundary coupling, four conditional bulk Hamiltonians give an exact time-dependent Schur, or generalized dephasing, channel. Its complete action is encoded in the output from the single probe preparation $|{+}{+}\rangle$. At every time where it is invertible, this channel has a canonical time-local master equation with at most three nonzero decoherence rates and diagonal Lindblad operators, all given in closed form by the coherence factors of the Schur multiplier. For exchange coupling, the available description gives the symmetry blocks rather than closed expressions for their coefficients. When the Hamiltonian conserves total magnetization and the bulk preparation commutes with bulk magnetization, the reduced superoperator and Choi matrix split into blocks of sizes $1, 4, 6, 4, 1$. Quantum-state texture is evaluated only on the output from $|{+}{+}\rangle$; exact relations to the Ising channel, its fidelities and its entanglement-generating ability follow from the Schur structure.

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1 Introduction

Two probes coupled to a common spin chain form an open quantum system whose reduced dynamics depends on the interaction at each contact. We consider Ising and exchange contacts separately, with no direct probe–probe interaction. The bulk starts in a fixed state, uncorrelated with the probe pair. Tracing out the bulk after unitary evolution defines a quantum channel on the pair.

Ising contacts conserve both probe z projections and produce generalized dephasing in the joint computational basis. Exchange contacts allow spin transfer between the probes and the bulk. Conservation of total magnetization then constrains the reduced channel when the bulk preparation respects the same rotation symmetry.

These two interactions lead to the following channel descriptions:

1. Ising coupling: an exact conditional-Hamiltonian reduction, an exact Schur channel, and reconstruction of that channel from the full output state for $|{+}{+}\rangle$.

2. Ising coupling, time-local description: the canonical form of the master equation, with its

decoherence rates, Lindblad operators and Hamiltonian expressed through the coherence factors.

3. Exchange coupling: exact total-magnetization sectors and, for a magnetization-invariant bulk preparation, a phase-covariant channel with five coherence-order blocks. These constraints leave the block entries to be computed from the many-body evolution.

4. Texture: exact grand-sum, return-probability and response-operator identities. For Ising coupling these yield channel fidelities and other properties without additional input preparations.

Section 2 fixes the model, the reduced channel and the texture conventions shared by both interactions. The two couplings are then treated in separate parts: Part I contains the Ising channel (Section 3) and its canonical form (Section 4); Part II contains the exchange channel (Section 5).

The derivations use exact finite-system dynamics, without a weak-coupling or Markov approximation. The channel structure fixes which probe operators can mix; their time dependence depends on the bulk Hamiltonian and its initial state.

2 Model, conventions and the reduced channel

2.1 Geometry and Hamiltonians

Let $N \geq 2$ be the number of *bulk* spins. There are $N+2$ spins in total, with physical order $A, 1, \ldots, N, B$. We use

$$S^\alpha = \sigma^\alpha/2, \qquad S^\pm = S^x \pm iS^y = (\sigma^x \pm i\sigma^y)/2, \qquad \hbar = 1. \tag{1}$$

The basis convention is $|0\rangle = |\uparrow\rangle$, $|1\rangle = |\downarrow\rangle$, and the probe basis is $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$. For formulas involving a product of probe and bulk operators, we group the Hilbert space as $\mathcal{H}_{AB} \otimes \mathcal{H}_E$ by a fixed permutation of the physical site ordering.

The reference bulk and the two alternative interactions are

$$H_E = J \sum_{j=1}^{N-1} (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y + \Delta S_j^z S_{j+1}^z), \tag{2}$$

$$V_{\mathrm{ZZ}} = g_A S_A^z S_1^z + g_B S_N^z S_B^z, \tag{3}$$

$$\begin{aligned} V_{\mathrm{ex}} &= g_A(S_A^x S_1^x + S_A^y S_1^y) + g_B(S_N^x S_B^x + S_N^y S_B^y) \\ &= \frac{g_A}{2}(S_A^+ S_1^- + S_A^- S_1^+) + \frac{g_B}{2}(S_N^+ S_B^- + S_N^- S_B^+). \end{aligned} \tag{4}$$

Thus $H^{(r)} = H_E + V_r$, with $r = \mathrm{ZZ}$ or ex. Identity operators on the remaining spins are implicit. The parameters J, g_A, g_B have energy units and Δ is dimensionless. Equal couplings mean $g_A = g_B = g$. The exchange flip-flop matrix element at the left contact is $g_A/2$ in this convention.

Equation (2) has no applied fields. Longitudinal bulk or probe fields preserve the magnetization arguments below, with their phases included in the conditional dynamics. Transverse *bulk* fields also preserve the Ising Schur reduction, but generally invalidate the exchange magnetization reduction because they break conservation of total z magnetization.

2.2 Product preparation and exact channel

Fix any bulk density matrix ω_E, independent of the probe input. The initial state and reduced evolution are

$$\rho(0) = \rho_{AB}(0) \otimes \omega_E, \tag{5}$$

$$U_t = e^{-iH^{(r)}t}, \qquad \Lambda_t(X) = \mathrm{Tr}_E[U_t(X \otimes \omega_E)U_t^\dagger], \tag{6}$$

$$\rho_{AB}(t) = \Lambda_t(\rho_{AB}(0)). \tag{7}$$

The product assumption separates the pair from the bulk; it imposes no factorization between A and B for the definition of the channel. Only the texture protocol fixes their input to $|{+}{+}\rangle$.

To see complete positivity and trace preservation directly, write $\omega_E = \sum_\nu p_\nu |\phi_\nu\rangle\langle\phi_\nu|$ and choose any orthonormal bulk output basis $\{|r\rangle\}$. Then

$$K_{r\nu}(t) = \sqrt{p_\nu}\, {}_E\langle r|U_t|\phi_\nu\rangle_E, \quad \Lambda_t(X) = \sum_{r,\nu} K_{r\nu} X K_{r\nu}^\dagger, \quad \sum_{r,\nu} K_{r\nu}^\dagger K_{r\nu} = I_4. \tag{8}$$

These are formal exact expressions in both regimes; evaluating them for an interacting bulk remains a many-body problem. Correlated probe–bulk preparations have reduced trajectories, but do not in general define the same CPTP map on arbitrary probe inputs.

2.3 Choi normalization

The *normalized* Choi state, with the output factor first, is

$$|\Phi_4\rangle = \frac{1}{2}\sum_{a=1}^{4} |a\rangle_{AB}|a\rangle_R, \qquad J_t = (\Lambda_t \otimes \mathrm{id}_R)(|\Phi_4\rangle\langle\Phi_4|). \tag{9}$$

It obeys $J_t \succeq 0$, $\mathrm{Tr}\, J_t = 1$ and $\mathrm{Tr}_{AB}\, J_t = I_R/4$. The corresponding unnormalized Choi matrix is $C_t = 4J_t$. The auxiliary system R represents a reference for the channel and is independent of the physical bulk.

2.4 Quantum-state texture of the $|{+}{+}\rangle$ preparation

Texture is defined relative to the probe computational basis. Its textureless state, and the sole probe preparation used for texture, is

$$|f_1\rangle = |{+}\rangle_A \otimes |{+}\rangle_B = \frac{1}{2}\sum_{a=1}^{4} |a\rangle, \qquad f_1 = |f_1\rangle\langle f_1|, \qquad (f_1)_{ab} = \frac{1}{4}. \tag{10}$$

Denote its full output state by $\rho_f(t) = \Lambda_t(f_1)$. For any probe state, the grand sum is real and satisfies

$$\Sigma(\rho) = \sum_{a,b} \rho_{ab} = 4\langle f_1|\rho|f_1\rangle, \qquad 0 \le \Sigma(\rho) \le 4. \tag{11}$$

For the single physical input $f_1 \otimes \omega_E$, set

$$p_f(t) = \langle f_1|\rho_f(t)|f_1\rangle = \Sigma(\rho_f(t))/4, \quad T_F(\rho_f) = 1 - p_f, \quad R(\rho_f) = -\ln p_f. \tag{12}$$

These are the fidelity texture and rugosity conventions used here [1, 2]. At $p_f = 0$, $R = +\infty$; initially $p_f = 1$ and both textures vanish. The texturizing powers evaluated on the unique textureless input are $\Omega_{T_F}(t) = T_F[\Lambda_t(f_1)]$ and $\Omega_R(t) = R[\Lambda_t(f_1)]$.

Since $4f_1 = (I + \sigma_A^x)(I + \sigma_B^x)$,

$$\Sigma(\rho_f) = 1 + \langle\sigma_A^x\rangle + \langle\sigma_B^x\rangle + \langle\sigma_A^x\sigma_B^x\rangle. \tag{13}$$

A joint x-basis measurement gives $p_f = P_{++}^{(x)}$ directly. Equivalently, apply a Hadamard to each probe and record outcome 00. This uses one measurement setting, whereas full channel reconstruction in the Ising case uses the additional coherence measurements described in Section 3.4.

For any channel, the dual map gives the grand-sum response operator

$$\mathcal{R}_t = 4\Lambda_t^*(f_1), \qquad \Sigma[\Lambda_t(X)] = \mathrm{Tr}(\mathcal{R}_t X). \tag{14}$$

These definitions apply in both coupling regimes. Their consequences for each interaction are derived in Sections 3.8 and 5.6.

Part I

Ising coupling

3 Ising coupling: exact generalized dephasing

3.1 Four conditional Hamiltonians

Since $[H^{(\mathrm{ZZ})}, S_A^z] = [H^{(\mathrm{ZZ})}, S_B^z] = 0$, label each probe basis state by $a = (s_A, s_B)$ with $s_A, s_B = \pm 1/2$, and set $P_a = |a\rangle\langle a|$. The Hamiltonian and propagator are

$$H^{(\mathrm{ZZ})} = \sum_a P_a \otimes H_a, \qquad H_a = H_E + g_A s_A S_1^z + g_B s_B S_N^z, \tag{15}$$

$$U_t = \sum_a P_a \otimes U_a(t), \qquad U_a(t) = e^{-iH_a t}. \tag{16}$$

This follows by inserting the spectral decompositions of S_A^z, S_B^z; orthogonal projectors exponentiate independently. It is exact for any bulk Hamiltonian acting only on E, including one with transverse fields.

For equal couplings, the four perturbations are

a	$H_a - H_E$
00	$\frac{g}{2}(S_1^z + S_N^z)$
01	$\frac{g}{2}(S_1^z - S_N^z)$
10	$-\frac{g}{2}(S_1^z - S_N^z)$
11	$-\frac{g}{2}(S_1^z + S_N^z)$

. (17)

These configurations label invariant branches. The input may still coherently occupy all four branches.

3.2 Schur form and its physical constraints

Let $E_{ab} = |a\rangle\langle b|$. Since $P_c E_{ab} P_d = \delta_{ca}\delta_{db} E_{ab}$, tracing the controlled propagator gives

$$\boxed{\Lambda_t(E_{ab}) = \gamma_{ab}(t) E_{ab}, \qquad \Lambda_t(X) = \Gamma(t) \circ X,} \tag{18}$$

where $\circ$ is the entrywise Schur product and

$$\gamma_{ab}(t) = \mathrm{Tr}_E[U_a(t)\omega_E U_b(t)^\dagger]. \tag{19}$$

We write $\Gamma(t) = (\gamma_{ab}(t))_{a,b=1}^{4}$: uppercase Γ denotes the Schur matrix, and lowercase γ_{ab} its complex, dimensionless entries. The channel itself is Λ_t. For a pure bulk, this is the complex conditional overlap $\langle\phi_0|U_b^\dagger U_a|\phi_0\rangle$. Its squared modulus is a conditional Loschmidt echo; the complex amplitude, including its phase, is needed for the channel.

The matrix Γ is a correlation matrix:

$$\Gamma = \Gamma^\dagger \succeq 0, \qquad \gamma_{aa} = 1, \qquad |\gamma_{ab}| \leq 1, \qquad \gamma_{ab}(0) = 1. \tag{20}$$

Indeed, setting $k_{r\nu,a} = \sqrt{p_\nu}\langle r|U_a|\phi_\nu\rangle$ gives $\gamma_{ab} = \sum_{r,\nu} k_{r\nu,a} k^*_{r\nu,b}$, a positive Gram factorization. The unit diagonal follows from unitarity and $\operatorname{Tr}\omega_E = 1$; 2×2 principal minors give the modulus bound. Conversely, every positive matrix with unit diagonal defines a CPTP Schur map. This converse characterizes the map class, not which members a specified XXZ Hamiltonian realizes at a given time.

All four populations are fixed and $\Lambda_t(I_4) = I_4$. Six complex off-diagonal factors specify the nontrivial action, subject to positivity. The common interacting bulk can correlate the two probes, so a product of independent single-probe dephasing channels cannot be assumed. There is no general monotonicity of $|\gamma_{ab}(t)|$.

3.3 Dephasing and depolarization

A depolarizing map has the form $\mathcal{D}_\lambda(X) = \lambda X + (1-\lambda)\operatorname{Tr}(X) I_4/4$. The Ising map fixes every P_a. Requiring $\mathcal{D}_\lambda(P_a) = P_a$ forces $\lambda = 1$. Thus a nonidentity Ising Schur map cannot be a depolarizing map. Dephasing includes coherent diagonal rotations and revivals; it does not imply irreversible or strictly dissipative evolution.

3.4 One output determines the complete channel

For the texture preparation f_1 of Eq. (10), with full output state $\rho_f(t) = \Lambda_t(f_1)$, the Schur form gives

$$\boxed{\rho_f(t) = \Gamma(t)/4, \qquad \Gamma(t) = 4\rho_f(t), \qquad \Lambda_t(X) = [4\rho_f(t)] \circ X.} \tag{21}$$

Consequently the *entire density matrix* of one known output determines every property of the instantaneous Ising channel. Temporal properties require these outputs at the relevant times. A single scalar texture value cannot replace this matrix.

The populations of ρ_f are $1/4$, so its 12 remaining real parameters are its six complex coherences. In a Pauli expansion, $\langle ZI\rangle = \langle IZ\rangle = \langle ZZ\rangle = 0$. The eight local measurement settings $(\alpha,\beta) \neq (z,z)$, $\alpha,\beta \in \{x,y,z\}$, recover the other Pauli coefficients, including single-spin expectations from their marginals. Thus one input preparation together with output tomography suffices.

3.5 Choi, Kraus and spectral consequences

Let $W : \mathcal{H}_{AB} \to \mathcal{H}_{AB} \otimes \mathcal{H}_R$ be the isometry $W|a\rangle = |aa\rangle$. Equations (9) and (18) give

$$J_t = \frac{1}{4}\sum_{a,b} \gamma_{ab}|aa\rangle\langle bb| = W\rho_f(t)W^\dagger. \tag{22}$$

The Choi support has dimension at most four. In particular,

$$\operatorname{rank} J_t = \operatorname{rank}\Gamma \leq 4, \tag{23}$$

$$\operatorname{spec}(J_t) = \operatorname{spec}(\Gamma/4) \text{ with twelve additional zeros}, \tag{24}$$

$$S(J_t) = S(\rho_f), \qquad \operatorname{Tr} J_t^2 = \operatorname{Tr}\rho_f^2 = \frac{1}{16}\sum_{a,b}|\gamma_{ab}|^2. \tag{25}$$

If $\Gamma = \sum_k \lambda_k v_k v_k^\dagger$ with normalized eigenvectors, one minimal Kraus family is

$$D_k = \sqrt{\lambda_k}\ \mathrm{diag}(v_{k,1}, \ldots, v_{k,4}), \qquad \Lambda_t(X) = \sum_k D_k X D_k^\dagger. \tag{26}$$

The superoperator is diagonal in the E_{ab} basis, with eigenvalues γ_{ab} and singular values $|\gamma_{ab}|$. Choi rank and superoperator rank are distinct: the identity has Choi rank one and superoperator rank sixteen.

The map is unitary iff $\operatorname{rank}\Gamma = 1$, in which case $\gamma_{ab} = u_a u_b^*$ with $|u_a| = 1$, giving the diagonal unitary $\mathrm{diag}(u_1, \ldots, u_4)$. This does not require that the unitary factor into two local rotations.

The partial transpose of J_t has eigenvalues $1/4$ (four times) and $\pm|\gamma_{ab}|/4$ for each $a < b$. Hence its reference–output negativity is

$$\mathcal{N}_{AB|R}(J_t) = \frac{1}{4}\sum_{a<b} |\gamma_{ab}|. \tag{27}$$

This is distinct from physical $A|B$ entanglement. The Ising map is entanglement breaking iff all off-diagonal γ_{ab} vanish: otherwise its Choi state has a negative partial transpose, while the diagonal case has the explicitly separable Choi state $\frac{1}{4}\sum_a |aa\rangle\langle aa|$.

3.6 Time-local dynamics and intermediate maps

The map is invertible iff every $\gamma_{ab}(t)$ is nonzero. At regular times its exact time-local generator obeys

$$\mathcal{L}_t(E_{ab}) = \frac{\dot\gamma_{ab}(t)}{\gamma_{ab}(t)} E_{ab}. \tag{28}$$

If Λ_s is invertible, the unique intermediate linear map is Schur multiplication by $Q_{ab}(t,s) = \gamma_{ab}(t)/\gamma_{ab}(s)$. Its diagonal is one, so

$$\Lambda_t \Lambda_s^{-1} \text{ is CPTP} \quad \Longleftrightarrow \quad Q(t,s) \succeq 0. \tag{29}$$

At singular times the division is undefined and this test cannot be applied. A vanished coherence factor that later becomes nonzero already precludes divisibility through that time, since a linear intermediate map cannot recover the annihilated matrix unit. Section 4 brings $\mathcal{L}_t$ to its canonical Lindblad-like form and gives the corresponding decoherence rates.

3.7 Probe entanglement from the same input

For two-qubit states, concurrence C and negativity $\mathcal{N}$ test entanglement of A with B; their definitions are in Appendix A. For a Schur channel, the output from $|{+}{+}\rangle$ also determines the maximum concurrence generated from separable probe states.

Proposition 3.1. *For a fixed Ising Schur channel, its maximum output concurrence over separable probe inputs is*

$$\max_{\rho \in \mathrm{Sep}(A:B)} C[\Lambda_t(\rho)] = C[\rho_f(t)] = C[\Gamma(t)/4]. \tag{30}$$

Proof. For a normalized pure product input $(a_0|0\rangle + a_1|1\rangle)\otimes(b_0|0\rangle + b_1|1\rangle)$, set $A = \sqrt{2}\,\mathrm{diag}(a_0, a_1)$ and $B = \sqrt{2}\,\mathrm{diag}(b_0, b_1)$. The output is $(A\otimes B)\rho_f(A\otimes B)^\dagger$ and has unit trace because ρ_f has uniform diagonal. The local-filter identity for two-qubit concurrence gives $C_{\mathrm{out}} = |\det A \det B| C(\rho_f) = 4|a_0 a_1 b_0 b_1| C(\rho_f) \le C(\rho_f)$. The identity follows from the spin-flip formula; singular filters follow by continuity. Convexity extends the bound to mixtures of product inputs. The input $|{+}{+}\rangle$ attains equality. □

In particular, the Ising map can create probe entanglement from some separable input iff $\rho_f(t)$ is entangled. Texture and Choi reference–output negativity are different quantities and do not replace this criterion.

3.8 Ising texture and channel fidelities

Using $\rho_f = \Gamma/4$ in Eq. (12) gives

$$p_f = \frac{1}{16}\sum_{a,b}\gamma_{ab} = \frac{1}{4} + \frac{1}{8}\sum_{a<b}\Re\gamma_{ab}, \tag{31}$$

$$\Omega_{T_F} = \frac{3}{4} - \frac{1}{8}\sum_{a<b}\Re\gamma_{ab}, \qquad \Omega_R = -\ln\left(\frac{1}{16}\sum_{a,b}\gamma_{ab}\right). \tag{32}$$

For the same Schur channel, the entanglement fidelity relative to the identity is $F_e = \langle\Phi_4|J_t|\Phi_4\rangle = p_f$. Therefore the average pure-input fidelity relative to the identity is

$$\boxed{F_e = p_f = \Sigma(\rho_f)/4, \qquad F_{\text{avg}} = \frac{4F_e + 1}{5} = \frac{1 + \Sigma(\rho_f)}{5}.} \tag{33}$$

The general relation between average and entanglement fidelities is standard [3]; $F_e = p_f$ uses the Schur structure. These are fidelities of the *two-probe map relative to the identity*, not fidelities of sending an unknown state from A to B.

Only the real sum of the six coherences enters p_f. It does not fix their individual magnitudes or phases and does not determine purity, probe entanglement, or CP divisibility. Coherent phase evolution can change texture even when the map is unitary.

3.9 Response operator and texture preservation

In the Ising case, comparing coefficients of X_{ab} in Eq. (14) yields

$$\mathcal{R}_t = \Gamma(t)^{\mathsf{T}}. \tag{34}$$

The transpose is ordinary, not a Hermitian conjugate. In general, fixing the free state, $\Lambda_t(f_1) = f_1$, and preserving the grand sum for every input, $\Lambda_t^*(f_1) = f_1$, are different conditions [2]. For this Schur family each condition forces $\gamma_{ab} = 1$ for all a, b, hence the identity channel. Consequently $\Omega_{T_F} = 0$ also certifies the identity within the known Ising family.

3.10 Summary of the Ising channel structure

Six complex overlaps between conditional bulk evolutions determine the Ising channel. The output $\rho_f = \Gamma/4$ therefore specifies its full action, including its Choi state, fidelities and ability to generate probe entanglement. Evaluating the overlaps requires the conditional dynamics for the chosen J, Δ, g_A, g_B, N and ω_E.

4 Ising coupling: canonical form of the time-dependent dephasing channel

The Schur structure of Eq. (18) fixes the time-local generator in closed form. Here we bring it to the canonical form of Hall *et al.* [4], which uniquely fixes the canonical rates. The associated

operators retain phase and degeneracy freedoms, as specified in Section 4.6. Every result of this section holds for an arbitrary fixed bulk preparation ω_E and applies to every probe input $\rho_{AB}(0)$, pure or mixed, product or entangled: the canonical form is a property of the channel Λ_t, which enters only through the coherence factors $\gamma_{ab}(t)$ of Eq. (19).

The notation distinguishes three quantities: $\gamma_{ab}(t)$ is a complex channel multiplier, $\kappa_{ab}(t)$ is its real logarithmic decay rate, and $\gamma_k(t)$ is a real canonical rate. The pair of configuration indices a, b labels a coherence; the single index k labels an eigenmode of the decoherence matrix. Both rates have units of inverse time, whereas Γ and γ_{ab} are dimensionless.

4.1 Time-local generator of the Schur channel

When the dynamical map is invertible, the reduced evolution obeys a time-local master equation with generator $\mathcal{L}_t = \dot{\Lambda}_t \Lambda_t^{-1}$ [4]. For the Schur channel this requires $\gamma_{ab}(t) \neq 0$ for all a, b (Section 3). Differentiating $\rho_{ab}(t) = \gamma_{ab}(t)\rho_{ab}(0)$ gives

$$\dot{\rho}_{ab}(t) = \ell_{ab}(t)\, \rho_{ab}(t), \qquad \ell_{ab}(t) = \frac{\dot{\gamma}_{ab}(t)}{\gamma_{ab}(t)} = \frac{d}{dt} \ln \gamma_{ab}(t), \tag{35}$$

so $\mathcal{L}_t(X)_{ab} = \ell_{ab}(t) X_{ab}$. The constraints of Eq. (20) imply $\ell_{aa} = 0$ and $\ell_{ba} = \ell_{ab}^*$. We separate its real and imaginary parts,

$$\ell_{ab} = -\kappa_{ab} + i\omega_{ab}, \qquad \kappa_{ab} = -\frac{d}{dt} \ln|\gamma_{ab}|, \qquad \omega_{ab} = \frac{d}{dt} \arg \gamma_{ab}. \tag{36}$$

Here $\kappa_{ab} = \kappa_{ba}$ and $\omega_{ab} = -\omega_{ba}$; the derivative ω_{ab} is taken on a locally continuous phase branch. Equivalently, $\omega_{ab} = \Im(\dot{\gamma}_{ab}/\gamma_{ab})$, which avoids artificial jumps of the principal argument. For any input with $\rho_{ab}(0) \neq 0$,

$$\kappa_{ab}(t) = -\frac{1}{|\rho_{ab}(t)|} \frac{d|\rho_{ab}(t)|}{dt}, \tag{37}$$

so κ_{ab} is the fractional rate of loss of the coherence between configurations a and b, independent of the input amplitude. Over an interval on which γ_{ab} does not vanish, $\int_{t_1}^{t_2} \kappa_{ab}\, dt = \ln(|\gamma_{ab}(t_1)|/|\gamma_{ab}(t_2)|)$.

4.2 Canonical form

Let $\{G_j\}_{j=1}^{15}$ be an orthonormal basis of traceless probe operators, $\mathrm{Tr}\, G_j = 0$ and $\mathrm{Tr}(G_j^\dagger G_k) = \delta_{jk}$. Any time-local master equation of a d-dimensional system can be written as [4]

$$\dot{\rho} = -i[H_{\mathrm{can}}(t), \rho] + \sum_{j,k} \mathsf{d}_{jk}(t) \Big(G_j \rho G_k^\dagger - \tfrac{1}{2}\{G_k^\dagger G_j, \rho\} \Big), \tag{38}$$

with a Hermitian traceless $H_{\mathrm{can}}(t)$ and a Hermitian decoherence matrix $\mathsf{d}(t)$. Here H_{can} acts on the probe pair and is distinct from the total probe–bulk Hamiltonian $H^{(\mathrm{ZZ})}$. Diagonalizing d gives

$$\dot{\rho} = -i[H_{\mathrm{can}}(t), \rho] + \sum_{k} \gamma_k(t) \Big(L_k(t) \rho L_k^\dagger(t) - \tfrac{1}{2}\{L_k^\dagger(t) L_k(t), \rho\} \Big), \tag{39}$$

where the $L_k(t)$ form an orthonormal basis of traceless operators. The canonical decoherence rates $\gamma_k(t)$ are the eigenvalues of $\mathsf{d}(t)$; they are uniquely determined, invariant under the choice of the basis $\{G_j\}$, time dependent, and can be negative [4]. We construct this form explicitly for $\mathcal{L}_t$ of Eq. (35).

4.3 Reduction to the diagonal operator subspace

Define the three diagonal, traceless and orthonormal operators

$$G_A = \frac{\sigma_A^z \otimes I}{2}, \qquad G_B = \frac{I \otimes \sigma_B^z}{2}, \qquad G_Z = \frac{\sigma_A^z \otimes \sigma_B^z}{2}. \tag{40}$$

For the configuration $a = (s_A, s_B)$ of Section 3, $G_j|a\rangle = g_j^a|a\rangle$ with the configuration vector

$$\mathbf{g}^a = (g_A^a, g_B^a, g_Z^a) = (s_A,\, s_B,\, 2s_As_B), \qquad \begin{array}{c|c} a & \mathbf{g}^a \\ \hline 00 & \frac{1}{2}(1,1,1) \\ 01 & \frac{1}{2}(1,-1,-1) \\ 10 & \frac{1}{2}(-1,1,-1) \\ 11 & \frac{1}{2}(-1,-1,1) \end{array}. \tag{41}$$

Complete $\{G_A, G_B, G_Z\}$ with any twelve orthonormal traceless off-diagonal operators. We take the ansatz of a diagonal Hamiltonian $H_{\text{can}} = \text{diag}(h_a)$ with $\sum_a h_a = 0$, and a decoherence matrix supported on the diagonal operators,

$$\mathsf{d} = S + iA, \qquad S = S^{\mathsf{T}} \in \mathbb{R}^{3\times 3}, \qquad A = -A^{\mathsf{T}} \in \mathbb{R}^{3\times 3}, \tag{42}$$

indexed by $j, k \in \{A, B, Z\}$, with all other entries zero.

For diagonal real G_j, the (a, b) entry of $G_j\rho G_k - \frac{1}{2}\{G_kG_j, \rho\}$ is $[g_j^a g_k^b - \frac{1}{2}(g_j^a g_k^a + g_j^b g_k^b)]\rho_{ab}$. Summing with $S_{jk} + iA_{jk}$, the symmetric part reduces to $-\frac{1}{2}(\mathbf{g}^a - \mathbf{g}^b)^{\mathsf{T}} S(\mathbf{g}^a - \mathbf{g}^b)$, and the antisymmetric part to $i\,\mathbf{g}^{a\mathsf{T}} A\,\mathbf{g}^b$, because $\mathbf{x}^{\mathsf{T}} A\mathbf{x} = 0$. The commutator contributes $-i(h_a - h_b)$. Hence the ansatz generates the entrywise action $\mathcal{L}_t(X)_{ab} = \ell_{ab}X_{ab}$ with

$$\ell_{ab} = -i(h_a - h_b) - \tfrac{1}{2}\Delta_{ab}^{\mathsf{T}} S\,\Delta_{ab} + i\,\mathbf{g}^{a\mathsf{T}} A\,\mathbf{g}^b, \qquad \Delta_{ab} = \mathbf{g}^a - \mathbf{g}^b. \tag{43}$$

Comparing with Eq. (36),

$$\kappa_{ab} = \tfrac{1}{2}\Delta_{ab}^{\mathsf{T}} S\,\Delta_{ab}, \tag{44}$$

$$\omega_{ab} = -(h_a - h_b) + \mathbf{g}^{a\mathsf{T}} A\,\mathbf{g}^b. \tag{45}$$

Sections 4.4 and 4.5 show that these equations have a unique solution (S, A, H_{can}) for every Hermitian ℓ with zero diagonal. The ansatz is then of the form (38), and uniqueness of the canonical form [4] identifies it as the canonical form of $\mathcal{L}_t$. Therefore:

Proposition 4.1. *At every time at which Λ_t is invertible, the canonical Hamiltonian of the Ising channel is diagonal in the configuration basis, and its decoherence matrix is supported on* $\text{span}\{G_A, G_B, G_Z\}$. *At most three canonical rates are nonzero; the remaining twelve vanish identically.*

No off-diagonal operator, and hence no operator that transfers population between configurations, appears in the generator. This is consistent with the fixed populations, $\gamma_{aa}(t) = 1$.

4.4 Dissipative part: the coherence magnitudes

The six difference vectors and Eq. (44) give

pair ab	Δ_{ab}	$\kappa_{ab} = \frac{1}{2}\Delta_{ab}^{\mathsf{T}} S \Delta_{ab}$
$00, 01$	$(0, 1, 1)$	$\frac{1}{2}(S_{BB} + S_{ZZ} + 2S_{BZ})$
$10, 11$	$(0, 1, -1)$	$\frac{1}{2}(S_{BB} + S_{ZZ} - 2S_{BZ})$
$00, 10$	$(1, 0, 1)$	$\frac{1}{2}(S_{AA} + S_{ZZ} + 2S_{AZ})$
$01, 11$	$(1, 0, -1)$	$\frac{1}{2}(S_{AA} + S_{ZZ} - 2S_{AZ})$
$00, 11$	$(1, 1, 0)$	$\frac{1}{2}(S_{AA} + S_{BB} + 2S_{AB})$
$01, 10$	$(1, -1, 0)$	$\frac{1}{2}(S_{AA} + S_{BB} - 2S_{AB})$

This linear map from the six entries of S to the six κ_{ab} is invertible. With the pair sums

$$K_A = \kappa_{00,10} + \kappa_{01,11}, \qquad K_B = \kappa_{00,01} + \kappa_{10,11}, \qquad K_{AB} = \kappa_{00,11} + \kappa_{01,10}, \tag{46}$$

for coherences in which only A, only B, or both probes flip, the inverse is

$$\begin{aligned} S_{AA} &= \tfrac{1}{2}(K_A + K_{AB} - K_B), & S_{AB} &= \tfrac{1}{2}(\kappa_{00,11} - \kappa_{01,10}), \\ S_{BB} &= \tfrac{1}{2}(K_B + K_{AB} - K_A), & S_{AZ} &= \tfrac{1}{2}(\kappa_{00,10} - \kappa_{01,11}), \\ S_{ZZ} &= \tfrac{1}{2}(K_A + K_B - K_{AB}), & S_{BZ} &= \tfrac{1}{2}(\kappa_{00,01} - \kappa_{10,11}). \end{aligned} \tag{47}$$

The entries have a direct reading:

- S_{AA} and S_{BB} describe dephasing of each probe along z.
- $S_{AB} > 0$ when the coherence between 00 and 11 decays faster than that between 01 and 10, as for noise acting equally on both probes; $S_{AB} < 0$ in the opposite, anticorrelated case.
- S_{AZ} (S_{BZ}) measures how the dephasing of A (B) depends on the configuration of the other probe.
- S_{ZZ} is the diagonal decoherence-matrix coefficient associated with $G_Z = \sigma_A^z \sigma_B^z/2$.

These entries are coefficients in the chosen operator basis, rather than canonical rates. They coincide with canonical rates when the full matrix $\mathsf{d} = S + iA$ is diagonal in that basis (Section 4.6). If Γ factorizes into two independent single-probe dephasing maps, $\gamma_{(s_A s_B),(s'_A s'_B)} = \alpha_{s_A s'_A} \beta_{s_B s'_B}$, then $\kappa_{00,10} = \kappa_{01,11} = \kappa^A$, $\kappa_{00,01} = \kappa_{10,11} = \kappa^B$ and $\kappa_{00,11} = \kappa_{01,10} = \kappa^A + \kappa^B$, so Eq. (47) gives $S = \mathrm{diag}(2\kappa^A, 2\kappa^B, 0)$. Nonzero S_{AB}, S_{AZ}, S_{BZ} or S_{ZZ} therefore signal dephasing correlated through the common bulk.

For $a \neq b$, each Δ_{ab} has squared norm 2. With the unit vector $\mathbf{e}_{ab} = \Delta_{ab}/\sqrt{2}$ and the normalized operator $G_{ab} = \sum_j (\mathbf{e}_{ab})_j G_j$,

$$\kappa_{ab} = \mathbf{e}_{ab}^{\mathsf{T}} S\, \mathbf{e}_{ab} = \mathbf{e}_{ab}^{\mathsf{T}} \mathsf{d}\, \mathbf{e}_{ab}, \tag{48}$$

since $\mathbf{e}^{\mathsf{T}} A \mathbf{e} = 0$. Each coherence decay rate is thus the decoherence matrix projected onto one fixed traceless diagonal operator; for example, $G_{01,10} = (\sigma_A^z - \sigma_B^z)/(2\sqrt{2})$ and $G_{00,11} = (\sigma_A^z + \sigma_B^z)/(2\sqrt{2})$.

4.5 Coherent part: the coherence phases

The energies h_a cancel from a sum of phase rates around a closed loop of three configurations. Write A through a vector $\mathbf{a}$, $A_{jk} = \epsilon_{jkl} a_l$, so that $\mathbf{x}^{\mathsf{T}} A \mathbf{y} = \mathbf{a} \cdot (\mathbf{x} \times \mathbf{y})$. Equation (45) then gives

$$C_{abc} \equiv \omega_{ab} + \omega_{bc} + \omega_{ca} = \mathbf{a} \cdot \left[(\mathbf{g}^b - \mathbf{g}^a) \times (\mathbf{g}^c - \mathbf{g}^a) \right]. \tag{49}$$

The four configurations have three independent loops,

$$\begin{aligned} C_1 &\equiv C_{00,01,10} = a_A + a_B - a_Z, \\ C_2 &\equiv C_{00,01,11} = -a_A + a_B - a_Z, \\ C_3 &\equiv C_{00,10,11} = -a_A + a_B + a_Z, \end{aligned} \tag{50}$$

which determine

$$a_A = \tfrac{1}{2}(C_1 - C_2), \qquad a_B = \tfrac{1}{2}(C_1 + C_3), \qquad a_Z = \tfrac{1}{2}(C_3 - C_2). \tag{51}$$

With A fixed, the remaining phase rates $\eta_{ab} = -\omega_{ab} + \mathbf{g}^{a\mathsf{T}} A\, \mathbf{g}^b$ satisfy $\eta_{ab} + \eta_{bc} + \eta_{ca} = 0$ around every loop, so $\eta_{ab} = h_a - h_b$ with

$$\boxed{H_{\mathrm{can}}(t) = \mathrm{diag}(h_a), \qquad h_a = \frac{1}{4} \sum_b \eta_{ab}.} \tag{52}$$

The Hamiltonian is a configuration-dependent energy shift of the probes induced by the bulk. The matrix A carries the part of the phase evolution that cannot be absorbed into four configuration energies; it vanishes iff all loop sums C_{abc} vanish.

4.6 Canonical rates and Lindblad operators

Let $\mathsf{d}(t)\mathbf{u}_k(t) = \gamma_k(t)\mathbf{u}_k(t)$, $k = 1, 2, 3$, with orthonormal eigenvectors $\mathbf{u}_k = (u_{Ak}, u_{Bk}, u_{Zk})$. Then

$$\boxed{\{\gamma_k(t)\}_{k=1}^{3} = \mathrm{spec}\big[S(t) + iA(t)\big], \qquad \gamma_4, \ldots, \gamma_{15} = 0,} \tag{53}$$

with S from Eq. (47) and A from Eq. (51). We order these three rates as $\gamma_1 \le \gamma_2 \le \gamma_3$. The additional twelve zeros are listed separately, so the indices $1, \ldots, 15$ do not constitute a globally sorted list. The canonical Lindblad operators are

$$\boxed{L_k(t) = \sum_{j\in\{A,B,Z\}} u_{jk}(t) G_j = \frac{1}{2}\big[u_{Ak}\sigma_A^z + u_{Bk}\sigma_B^z + u_{Zk}\sigma_A^z\sigma_B^z\big],} \tag{54}$$

which are traceless and satisfy $\mathrm{Tr}(L_j^\dagger L_k) = \delta_{jk}$. They are diagonal in the configuration basis,

$$L_k(t)|a\rangle = (\mathbf{u}_k(t)\cdot\mathbf{g}^a)|a\rangle, \tag{55}$$

so each L_k imprints a configuration-dependent amplitude and phase. If $A = 0$, the eigenvectors can be chosen real and the L_k are Hermitian; for $A \neq 0$ they are diagonal but generally non-Hermitian. For nondegenerate rates each L_k is unique up to a phase; within a degenerate rate subspace, orthonormal eigenoperators may be mixed by a unitary transformation [4]. Ordering rates at each time does not track an eigenoperator continuously through a crossing.

The canonical master equation of the Ising channel is therefore

$$\dot\rho = -i[H_{\mathrm{can}}(t), \rho] + \sum_{k=1}^{3} \gamma_k(t)\Big(L_k(t)\rho L_k^\dagger(t) - \tfrac{1}{2}\{L_k^\dagger(t)L_k(t), \rho\}\Big), \tag{56}$$

with H_{can}, γ_k and L_k determined by $\Gamma(t)$ alone. All operators in it are diagonal and mutually commute, so it fixes the populations and is unital. The L_k are not raising or lowering operators: such operators would move population, which Proposition 4.1 excludes for this channel.

4.7 Sign of the rates and CP divisibility

Following Hall *et al.* [4], a time-local master equation is called Markovian at time t iff all canonical rates are nonnegative, and non-Markovian iff at least one rate is strictly negative. On an interval where the map is differentiable and invertible, nonnegative rates throughout the interval are equivalent to CP divisibility [4]. A negative rate does not invalidate Eq. (56), and Λ_t remains CPTP; it signals that the intermediate maps near t are not completely positive, consistent with the finite-interval test of Eq. (29).

For the Ising channel the sign structure follows from S and A: write $\gamma_{\min} = \gamma_1$ and $\gamma_{\max} = \gamma_3$ for the extreme eigenvalues of the 3×3 diagonal-operator block.

1. The smallest rate obeys $\gamma_{\min} \le \lambda_{\min}(S)$. Indeed $S = (\mathsf{d} + \mathsf{d}^{\mathsf{T}})/2$, and d^{T} has the same spectrum as d, so Weyl's inequality gives $\lambda_{\min}(S) \ge \gamma_{\min}$. A non-positive-semidefinite S therefore implies a negative canonical rate; the phase part A cannot restore positivity.

2. By Eq. (48), for every distinct pair $a \neq b$,

$$\gamma_{\min}(t) \leq \kappa_{ab}(t) \leq \gamma_{\max}(t). \tag{57}$$

Hence a negative rate occurs whenever any coherence magnitude $|\gamma_{ab}|$ increases. Likewise $\gamma_{\min} \leq S_{jj}$ for $j \in \{A, B, Z\}$; in particular $K_{AB} > K_A + K_B$, i.e. $S_{ZZ} < 0$, implies a negative rate even if every κ_{ab} is positive.

3. $S \succeq 0$ iff there is a real 3×3 matrix M with $\kappa_{ab} = \frac{1}{2}|M\Delta_{ab}|^2$ for all pairs. In that case, the six decay rates are one half of squared Euclidean distances between four points $M\mathbf{g}^a$ representing the configurations.

The complete condition is $\mathsf{d} = S + iA \succeq 0$. Although $S \succeq 0$ and nonnegative κ_{ab} are necessary for this condition, neither alone is sufficient. For example, $S = I_3$ with $A_{AB} = -A_{BA} = 2$ and all other entries of A zero has $\kappa_{ab} = 1$ for all $a \neq b$, but $\mathrm{spec}(S + iA) = \{-1, 1, 3\}$.

4.8 Singular times

The rates are built from the logarithmic derivatives $\ell_{ab} = \dot{\gamma}_{ab}/\gamma_{ab}$. The inverse-map construction fails when a coherence factor vanishes [4]. For the finite system with a time-independent Hamiltonian, each $\gamma_{ab}(t)$ is analytic. Near a zero of order $r \geq 1$ at t_*,

$$\gamma_{ab}(t) = c(t - t_*)^r[1 + O(t - t_*)], \quad c \neq 0, \qquad \ell_{ab}(t) = \frac{r}{t - t_*} + O(1). \tag{58}$$

Thus $\kappa_{ab} = -r/(t - t_*) + O(1)$ has a pole, and at least one canonical rate becomes unbounded by Eq. (57). The phase rate ω_{ab} need not diverge: the leading pole in Eq. (58) is real. At t_* the rates defined by $\dot{\Lambda}_t \Lambda_t^{-1}$ are undefined. The time-local form applies on neighboring intervals where all coherence factors are nonzero. The state and its derivative remain bounded: $\rho_{ab}(t) = \gamma_{ab}(t)\rho_{ab}(0)$ and its time derivative stay bounded, with $|\dot{\rho}_{ab}| \leq |\dot{\gamma}_{ab}||\rho_{ab}(0)|$. According to Eq. (37), κ_{ab} measures a fractional rather than an absolute change. Its signed integral is finite on a closed interval containing no zeros, as shown below Eq. (37). At an exact zero the corresponding one-sided logarithmic integrals diverge; a principal-value cancellation does not define a finite ordinary integral across the singularity.

Part II

Exchange coupling

5 Exchange coupling: exact symmetry reduction

5.1 Total charge and Hamiltonian sectors

For the field-free model define

$$M_S = S_A^z + S_B^z, \qquad M_E = \sum_{j=1}^{N} S_j^z, \qquad M = M_S + M_E. \tag{59}$$

Each exchange term transfers one unit between a probe and the bulk, so $[H^{(\mathrm{ex})}, M] = 0$, while individual probe S^z values generally are not conserved. The total-charge sector m has dimension

$$\dim \mathcal{H}_m = \binom{N+2}{(N+2)/2 - m}. \tag{60}$$

This Hamiltonian decomposition requires no assumption on the input; coherences between occupied sectors must still be retained. There is no four-branch controlled propagator of the form (15) for exchange coupling.

5.2 Bulk assumption and proof of phase covariance

For the channel reduction assume

$$[\omega_E, M_E] = 0. \tag{61}$$

Support in a single well-defined bulk magnetization sector is sufficient. An incoherent mixture of sectors also suffices. A coherent superposition of different bulk magnetizations generally does not.

Let $V_\theta = e^{-i\theta M_S}$ and $W_\theta = e^{-i\theta M_E}$. Conservation gives $[U_t, V_\theta \otimes W_\theta] = 0$, while (61) gives $W_\theta \omega_E W_\theta^\dagger = \omega_E$. Inserting these into the partial trace proves

$$\boxed{\Lambda_t(V_\theta X V_\theta^\dagger) = V_\theta \Lambda_t(X) V_\theta^\dagger.} \tag{62}$$

This is $U(1)$ phase covariance. It permits exchange of magnetization with the bulk; it does not imply $\Lambda_t^*(M_S) = M_S$ or unitality.

5.3 Five superoperator blocks

The probe charges are $(m_{00}, m_{01}, m_{10}, m_{11}) = (1, 0, 0, -1)$. The operator E_{ab} has coherence order $q = m_a - m_b$, meaning $[M_S, E_{ab}] = qE_{ab}$. Under a rotation it acquires the phase $e^{-iq\theta}$, so covariance implies

$$\Lambda_t(\mathcal{V}_q) \subseteq \mathcal{V}_q, \qquad \mathcal{V}_q = \mathrm{span}\{E_{ab} : m_a - m_b = q\}. \tag{63}$$

The ordered bases $\mathcal{B}_q$ of these spaces are

q	Dimension	Basis of $\mathcal{V}_q$
0	6	$E_{00,00}, E_{01,01}, E_{10,10}, E_{11,11}, E_{01,10}, E_{10,01}$
$+1$	4	$E_{00,01}, E_{00,10}, E_{01,11}, E_{10,11}$
-1	4	Adjoints of the $q = +1$ basis
$+2$	1	$E_{00,11}$
-2	1	$E_{11,00}$

Thus the 16×16 superoperator has the exact reduction

$$\boxed{\Lambda_t \simeq T^{(-2)} \oplus T^{(-1)} \oplus T^{(0)} \oplus T^{(+1)} \oplus T^{(+2)}, \quad (d_q) = (1, 4, 6, 4, 1).} \tag{64}$$

Operators of the same order may mix. Hermiticity relates the opposite orders. In the ordered basis $\mathcal{B}_q$, the exact entries are

$$T^{(q)}_{cd,ab}(t) = \sum_{r,\nu} (K_{r\nu})_{ca} (K_{r\nu})^*_{db}, \qquad E_{ab}, E_{cd} \in \mathcal{B}_q. \tag{65}$$

Equation (65) is a formal construction, not a closed evaluation of these functions for general interacting XXZ dynamics.

The normalized Choi state satisfies the equivalent symmetry condition

$$[J_t, M_S \otimes I_R - I_{AB} \otimes M_S^{\mathsf{T}}] = 0. \tag{66}$$

Grouping output–reference kets $|c\rangle|a\rangle$ by $m_c - m_a$ gives Choi blocks of the same sizes $1, 4, 6, 4, 1$. Choi blocks and superoperator blocks are different rearrangements of channel coefficients and should not be identified as matrices or assigned the same spectra.

5.4 Kraus operators with definite charge transfer

Under (61), diagonalize ω_E within its charge sectors: $M_E|\phi_\nu\rangle = \mu_\nu|\phi_\nu\rangle$. Choose $M_E|r\rangle = \mu_r|r\rangle$. Conservation of the total input and output charge in Eq. (8) yields

$$(K_{r\nu})_{ca} \neq 0 \implies m_c - m_a = \mu_\nu - \mu_r, \qquad [M_S, K_{r\nu}] = (\mu_\nu - \mu_r)K_{r\nu}. \tag{67}$$

Only transfers between -2 and $+2$ can act on the two probes. The transfer label belongs to a Kraus operator; coherence order labels an input or output matrix unit. Their roles in the Choi and superoperator reductions are related but distinct.

5.5 Reduced states at fixed total magnetization

If the *global* initial state is supported in one total-M sector, the reduced probe state has the exact form

$$\rho_{AB}(t) = \begin{pmatrix} p_{00} & 0 & 0 & 0 \\ 0 & p_{01} & c & 0 \\ 0 & c^* & p_{10} & 0 \\ 0 & 0 & 0 & p_{11} \end{pmatrix}. \tag{68}$$

For a pure global ket, resolve its four probe components. Their bulk kets have charges $m-1, m, m, m+1$. Orthogonality of different bulk charges eliminates every coherence except $01 \leftrightarrow 10$. Linearity extends the conclusion to mixtures within the sector. The populations and c remain time dependent, and the concurrence is

$$C = 2\max\{0, |c| - \sqrt{p_{00}p_{11}}\}. \tag{69}$$

This restriction does *not* apply to the texture protocol merely because the bulk has fixed magnetization. The input $|{+}{+}\rangle$ contains probe charges $+1, 0, -1$ with probabilities $1/4, 1/2, 1/4$, and hence occupies three total-charge sectors for a fixed-charge bulk. Their coherences cannot be discarded. In particular, f_1 itself is not of the form (68).

Reflection and global spin flip give additional covariance only when both the Hamiltonian and bulk preparation have the required symmetry. Equal contacts or a symmetric Hamiltonian alone do not impose that symmetry on an arbitrary reduced trajectory.

5.6 Exchange texture and the limits of one input

Equations (12), (13) and (14) hold for exchange coupling as well. However, ρ_f is generally not a Schur multiplier matrix and does not determine the exchange channel. Under Eq. (61), f_1 has a nonzero component in every $\mathcal{V}_q$, but one output determines only each block's action on that particular component. It cannot determine a general 6×6 or 4×4 block.

There is one exact scalar exception. Since $\mathcal{V}_{+2}$ is one-dimensional, write $\Lambda_t(E_{00,11}) = \eta_{+2}(t)E_{00,11}$. Then

$$\eta_{+2}(t) = 4[\rho_f(t)]_{00,11}, \qquad \eta_{-2}(t) = \eta_{+2}(t)^*. \tag{70}$$

This inference requires covariance; it is not guaranteed for a bulk coherent across magnetization sectors or for transverse-field dynamics. The equality $F_e = p_f$ is likewise not a general exchange-channel identity, although $F_{\text{avg}} = (4F_e + 1)/5$ still holds.

5.7 Summary of the exchange channel structure

For exchange coupling, conservation of total magnetization and a magnetization-invariant bulk preparation give five coherence-order blocks. Symmetry fixes their dimensions and forbids mixing

between different orders, but leaves the entries within each block to be calculated. The output from $|{+}{+}\rangle$ determines the texture and the one-dimensional highest-order response; it does not specify the full exchange channel. A general closed expression for the interacting exchange coefficients remains unavailable.

A State diagnostics and their interpretation

For $\rho = \rho_{AB}$, the state purity and entropy are

$$P = \operatorname{Tr} \rho^2, \qquad S(\rho) = -\sum_k \lambda_k \ln \lambda_k, \qquad 0 \ln 0 = 0. \tag{71}$$

Logarithms are natural. The probe mutual information is $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho)$, with $\rho_A = \operatorname{Tr}_B \rho$ and $\rho_B = \operatorname{Tr}_A \rho$. It includes classical and quantum correlations.

Probe negativity and concurrence are

$$\mathcal{N}_{A|B}(\rho) = \frac{\|\rho^{T_B}\|_1 - 1}{2}, \tag{72}$$

$$\widetilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \qquad C(\rho) = \max(0, r_1 - r_2 - r_3 - r_4), \tag{73}$$

where the r_i are the decreasing nonnegative square roots of the eigenvalues of $\rho\widetilde{\rho}$ [5]. For two qubits, $C > 0$ and $\mathcal{N} > 0$ are equivalent tests of entanglement. They concern $A|B$, while $S(\rho_{AB})$ concerns the pair's marginal mixedness. Only for a globally pure state does $S(\rho_{AB})$ equal entanglement entropy across $AB|E$.

For the Ising texture input,

$$P_f(t) = \frac{1}{4} + \frac{1}{8}\sum_{a<b} |\gamma_{ab}(t)|^2, \qquad \frac{1}{4} \le P_f \le 1. \tag{74}$$

Its local probe coherences are

$$(\rho_A)_{01} = \frac{\gamma_{00,10} + \gamma_{01,11}}{4}, \qquad (\rho_B)_{01} = \frac{\gamma_{00,01} + \gamma_{10,11}}{4}, \tag{75}$$

and each local diagonal is $(1/2, 1/2)$. For a pure bulk, the global state is $\frac{1}{2}\sum_a |a\rangle \otimes U_a|\phi_0\rangle$; it is a product across $AB|E$ iff its four conditional bulk kets are proportional. For a mixed bulk, reduced entropy or purity alone does not quantify entanglement across that cut.

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